

Abhay Singh Patel

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SUMMARY

Full-Stack Software Engineer specializing in **distributed backend systems, AI-powered applications, and robotics simulation infrastructure**. Experienced in building production systems using **Django, Next.js, and Docker** while designing **CI/CD pipelines and automated cloud infrastructure** using **Terraform and Ansible**. Proven expertise in developing **Omniverse Isaac Sim extensions, RAG-based AI systems, and asynchronous task queues** for scalable AI inference workloads. Track record of improving system performance through **Redis caching, GPU-based infrastructure automation, and microservices architecture**.

EDUCATION

2020 – 2024 **Indian Institute of Technology (IIT) Bhilai**, B.Tech in Computer Science Raipur, Chhattisgarh, India
2017 – 2019 **Central Academy**, Intermediate Kota, Rajasthan

WORK EXPERIENCE

Software Engineer | AuraML Apr 2025 – Present | Bangalore

- Developed an **asynchronous task queue system for Image-to-3D processing APIs**, enabling background job execution with real-time polling updates in the frontend.
- Automated **GPU instance provisioning for robotics simulations** using **Terraform and Ansible**, dynamically creating and terminating compute resources for simulation workloads.
- Built microservices including a **Rust-based storage service** and a **WebRTC communication service**, integrated via **API Gateway and Nginx reverse proxy**.
- Built custom **Omniverse Isaac Sim extensions** using the **Omniverse SDK** and developed a custom **Isaac Kit application** for scene configuration, robot configuration, and simulation workflow management.
- Integrated custom **Omniverse Isaac Sim extensions** with **Text-to-3D pipelines**, enabling automated simulation asset generation, ROS2-based robot communication, and distributed runtime connectivity using **Zenoh routers** for robotics simulation workflows.
- Designed and implemented a **centralized CI/CD pipeline with automated deployments** using **GitHub Actions**, improving deployment reliability across multiple microservices.
- Implemented **centralized authentication and authorization using Keycloak with OAuth 2.0**, enabling secure **Single Sign-On (SSO)** via Google and GitHub across internal platforms.

Tech: *Terraform, Ansible, Rust, C++, Python, Django, Next.js, Docker, GitHub Actions, Nginx, WebRTC, Microservices, Omniverse SDK, Isaac Sim, ROS2, Zenoh, OAuth 2.0, Keycloak*

Software Engineer | IVY Homes Nov 2024 – Apr 2025 | Bangalore

- Implemented **Redis caching for CMS APIs**, reducing data retrieval latency by **50%** and improving overall platform responsiveness.
- Built backend APIs and automation workflows integrated with **Zoho Flow, WATI, and Zepto Mail**, enabling automated communication workflows that handled **70% of customer messaging operations**.
- Designed and deployed **data scraping pipelines on GCP**, automating ingestion and processing of large-scale property datasets.
- Implemented **GCP Pub/Sub messaging** to support **real-time communication across microservices**, improving reliability of asynchronous data processing.
- Designed scalable APIs and optimized **PostgreSQL database schemas**, improving backend query efficiency for frontend modules built in **Next.js**.

Tech: *GCP, Redis, PostgreSQL, Django, Next.js, Pub/Sub, Zoho Flow, WATI, Zepto Mail, Data Pipelines, Python*

Software Engineer | MAQ Software

May 2024 – Oct 2024 | Hyderabad, India

- Worked on **Azure** Data Lakehouse for large-scale data storage and analytics.
- Cleaned and transformed datasets using **PySpark**.
- Developed Power BI dashboards and custom visualization cards for analytics.

Tech: *Azure, PySpark, Power BI, Data Lakehouse, Python*

INTERNSHIPS

Web Developer Intern | Param Innovations

May – July 2023 | Bangalore, India

- Built frontend interfaces using **Chakra UI** and **React** hooks for accessible design.
- Developed responsive static websites and optimized state management for faster performance and SEO.

Tech: *React, Chakra UI, JavaScript, SEO Optimization*

Runner-up – Smart India Hackathon | NTRO

Aug 2022 | Remote

- Developed an end-to-end prototype for ransomware detection in university systems during a 24-hour hackathon.

Developer Intern | Agronomist Agrotech (College Startup)

Jan – Mar 2023 | Remote

- Worked on **Raspberry Pi** functionalities and enhancements.
- Contributed to the web interface using **Next.js**.

OPEN SOURCE CONTRIBUTIONS

Contributor | Twenty CRM (Open-Source CRM)

[Merged PR](#)

- Contributed to the open-source **Twenty CRM** platform built with **TypeScript, React, NestJS, PostgreSQL, and Redis**.
- Developed and merged UI improvements for navigation overflow handling and tooltip-based accessibility enhancements.
- Collaborated with maintainers through code reviews, debugging discussions, and CI validation workflows.

Tech: *TypeScript, React, NestJS, PostgreSQL, Redis, Open Source*

PROJECTS

Atomic – Perplexity-style AI Search Platform

[Live Demo](#) | [GitHub](#)

- Built an AI-powered search platform integrating **SearXNG** with LLMs for contextual answer generation and web-augmented search.
- Designed a scalable **asynchronous task pipeline** using **Celery** and **RabbitMQ** for handling long-running AI inference workloads.
- Developed backend APIs using **Django REST Framework** and frontend using **Next.js**.
- Implemented containerized deployment using **Docker** with **PostgreSQL** for persistent storage and CI/CD pipelines for automated deployments.

Tech: *Django REST Framework, Next.js, PostgreSQL, Celery, RabbitMQ, Docker, CI/CD, SearXNG, LLM*

- Built a document understanding platform enabling users to upload books and perform contextual semantic search and Q&A using **RAG pipelines**.
- Implemented semantic chunking using **semantic-chunking** and generated embeddings using a self-hosted **bge-small-en** model with vector storage in **Qdrant**.
- Integrated **Mistral API** and local LLM inference using **Ollama** for contextual retrieval, summarization, and document-based question answering.
- Developed asynchronous workflows using **Celery** and **RabbitMQ** for scalable document ingestion and embedding generation.

Tech: *Django REST Framework, PostgreSQL, Celery, RabbitMQ, Qdrant, Mistral LLM, bge-small-en, RAG, Docker*

- Built a video understanding platform to process YouTube transcripts and generate contextual summaries and Q&A using **RAG pipelines**.
- Implemented semantic chunking using **semantic-chunking** and generated embeddings using a self-hosted **bge-small-en** model with vector retrieval in **Qdrant**.
- Designed retrieval pipelines with query expansion and multi-stage summarization for improved response accuracy.
- Integrated **Mistral API** and local LLM inference using **Ollama** with **8K token context windows** for transcript summarization and contextual video Q&A.
- Developed asynchronous workflows using **Celery** and **RabbitMQ** for scalable transcript processing and embedding generation.

Tech: *Django REST Framework, PostgreSQL, Celery, RabbitMQ, Qdrant, bge-small-en, RAG, Docker*

SKILLS

Languages	JavaScript, Python, C++, Rust, SQL, TypeScript
Backend	Django, Django REST Framework, Node.js (Express), NestJS, Microservices Architecture
Frontend	Next.js, React.js, Chakra UI, Shadcn, MUI UI
DevOps & Cloud	Docker, Kubernetes, Terraform, Ansible, GitHub Actions, Nginx, Linux, GCP (Pub/Sub, Compute Engine), Azure (Data Lakehouse)
Databases & Caching	PostgreSQL, Redis, Qdrant (Vector DB), MongoDB
AI/ML & Data	RAG Pipelines, LLMs (Mistral, Ollama), Celery, RabbitMQ, PySpark, semantic-chunking, bge-small-en embeddings
Robotics & Simulation	Omniverse Isaac Sim, Omniverse SDK, ROS2, Zenoh, WebRTC
APIs & Tools	REST APIs, GraphQL, OAuth 2.0, Keycloak, API Gateway, CI/CD Pipelines, Async Task Queues, SearXNG
Integration & Automation	Zoho Flow, WATI, Zepto Mail, Pub/Sub Messaging, Data Scraping Pipelines

ACHIEVEMENTS

- **Expert (1640)** on Codeforces and **4 (1783)** on CodeChef.
- **Rank 1** in college and **1091 rank** in CodeKaze'23.
- **Runner-up** at Smart India Hackathon 2024.
- Captain of district cricket team; Defender of district football team.
- Global Rank **1544** in Google Kickstart (Round E).